

GAYA, Niger

WMO: 610990

Lat: 11.883N

Lon: 3.450E

Elev: 202

StdP: 98.92

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.8	19.8	-7.3	2.1	28.0	-5.9	2.4	28.1	9.2	23.1	8.3	23.4	3.1	40	0.398

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	11.7	41.4	21.9	40.6	22.1	39.7	22.2	27.5	33.8	27.2	33.5	26.8	33.1	2.7	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.1	22.0	30.3	25.7	21.5	30.0	25.3	21.0	29.7	88.8	34.2	87.1	33.7	85.6	33.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.4	6.3	5.4	15.7	43.1	2.3	0.9	14.1	43.8	12.7	44.3	11.5	44.8	9.8	45.4		
			7.5	28.8	1.4	0.7	6.5	29.3	5.7	29.7	5.0	30.1	4.0	30.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.7	26.8	29.7	33.2	34.5	32.6	30.0	28.1	27.0	27.9	29.8	29.5	27.3
	DBStd	3.08	2.24	2.65	1.91	1.72	2.29	2.11	1.76	1.47	1.58	1.32	1.28	1.87
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	7187	522	553	718	734	701	599	561	527	537	615	586	535
	CDD18.3	4146	263	319	460	484	442	349	302	269	287	357	336	277
	CDH23.3	54803	2882	4301	7248	7856	6711	4673	3383	2558	3119	4536	4439	3097
	CDH26.7	31264	1455	2522	4844	5474	4316	2521	1429	859	1290	2394	2571	1589
Wind	WSAvg	2.3	3.5	3.2	2.9	2.4	2.4	2.0	1.7	1.6	1.5	1.5	1.9	3.0
Precipitation	PrecAvg	791	0	2	3	16	67	113	180	227	163	18	0	0
	PrecMax	1038	0	44	32	59	213	210	350	337	252	58	2	6
	PrecMin	472	0	0	0	0	5	42	59	126	34	0	0	0
	PrecStd	111	0	8	7	18	41	46	67	65	51	19	0	1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.7	40.6	42.0	42.6	41.5	38.6	36.1	33.7	35.1	37.9	38.7	37.5
		MCWB	18.6	20.1	20.7	22.4	24.4	25.6	26.1	26.4	27.0	23.0	21.1	19.3
	2%	DB	36.1	39.2	41.0	41.7	40.2	37.2	34.2	32.5	34.0	36.8	37.7	36.4
		MCWB	17.6	19.7	20.4	22.2	25.0	25.7	25.9	26.2	26.7	23.8	21.0	18.7
	5%	DB	34.8	37.9	40.2	41.0	39.0	36.1	33.2	31.6	33.1	35.9	36.9	35.2
		MCWB	17.1	18.9	20.3	22.5	25.4	25.7	25.9	26.0	26.5	24.4	20.6	18.2
	10%	DB	33.4	36.4	39.3	40.1	37.9	34.9	32.2	30.7	32.1	34.9	36.0	33.8
		MCWB	16.5	18.3	20.0	23.1	25.5	25.8	25.8	25.8	26.2	25.1	20.0	17.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.6	24.8	25.9	27.1	27.7	27.7	27.4	27.2	27.8	28.0	25.5	21.1
		MCDB	32.1	34.6	36.9	37.5	35.9	34.2	32.8	31.9	33.1	33.9	33.4	34.8
	2%	WB	19.3	22.8	24.7	26.5	27.2	27.1	26.8	26.7	27.3	27.4	24.4	19.9
		MCDB	33.8	32.5	35.7	36.9	35.6	33.7	32.0	31.0	32.4	33.4	32.4	34.1
	5%	WB	17.9	20.7	23.9	26.0	26.7	26.7	26.5	26.3	26.9	26.9	23.6	19.1
		MCDB	33.6	35.7	34.8	36.1	35.1	33.2	31.5	30.5	31.8	32.9	31.5	33.3
	10%	WB	16.8	19.1	22.9	25.5	26.4	26.3	26.1	26.0	26.5	26.5	22.6	18.2
		MCDB	32.7	35.9	34.6	35.5	34.6	32.6	31.0	29.9	31.1	32.2	30.9	32.5
Mean Daily Temperature Range		MDBR	13.2	12.9	12.6	11.7	10.4	9.3	8.0	7.1	8.2	10.9	14.4	13.5
	5% DB	MCDBR	14.7	14.1	13.4	12.6	11.5	10.6	9.3	8.4	9.6	12.4	15.1	15.1
		MCWBR	6.3	6.1	5.9	4.4	3.1	2.9	2.8	3.0	3.8	3.4	4.5	5.9
	5% WB	MCDBR	14.2	12.8	11.8	11.2	10.2	9.4	8.6	7.8	9.1	10.3	13.4	14.6
		MCWBR	6.8	6.1	5.9	3.5	3.1	3.2	3.2	3.3	3.9	3.7	4.4	6.1
Clear-Sky Solar Irradiance	taub	0.637	0.710	0.776	0.745	0.679	0.628	0.568	0.550	0.557	0.621	0.569	0.552	
	taud	1.673	1.518	1.426	1.555	1.666	1.784	1.970	2.009	1.973	1.790	1.805	1.823	
	Ebn at Noon	685	655	625	643	674	700	746	771	771	712	734	740	
	Edh at Noon	243	294	327	286	251	220	184	179	187	221	211	205	
All-Sky Solar Radiation	RadAvg	5.44	5.74	6.01	6.19	6.00	5.72	5.37	5.00	5.54	5.94	5.74	5.44	
	RadStd	0.26	0.25	0.31	0.22	0.18	0.23	0.21	0.24	0.23	0.19	0.14	0.25	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page